

# When Does Density Beat Centroids?

## A Comparative Study of Clustering Methods Across Data Geometries

[DECIDE: final title]

SC4020 Project 1 — Group XX

[All: full names exactly as on student cards; replace XX with group ID]

### Abstract

Owner: Ky (write last)

[Ky: 150–250 words: task, methods compared, datasets, 3 headline findings with numbers, one-line conclusion.]

[DECIDE: Thesis. Proposed: the geometry of the data (cluster shape, density variation, dimensionality, gaps between groups) decides which method works, and internal scores such as silhouette cannot tell you which. Every member’s data supports this (Section 3.4.5).]

## 1 Introduction

Owner: Wei Yew (draft); Ky to review

### 1.1 Motivation

Clustering is used where no answer key exists. A seismologist grouping earthquake epicentres wants to find active fault systems. A housing analyst grouping resale transactions wants market segments. Neither has labels to check a result against. Any algorithm they run will still return a partition, along with a score.

The methods differ in what they take a cluster to be. K-means [12] looks for compact groups around centroids and assigns every point to one of them. DBSCAN [8] looks for dense regions separated by sparse space and leaves isolated points as noise. Gaussian mixtures stretch the K-means model to ellipses. Spectral clustering drops the shape assumption and works on a similarity graph. When a method’s assumption matches the data, it recovers real structure. When it does not, it still returns clusters. On one year of global earthquakes, K-means places five plate-boundary belts, lying 2,932 to 11,449 km apart, in a single cluster (Section 3.4.1).

The practical question is which method fits which data. It has to be answered without labels.

### 1.2 Problem statement

Without labels, methods and parameters are usually chosen by an internal index. The most common is the silhouette coefficient [16]. It compares each point’s mean distance to its own cluster with its mean distance to the nearest other cluster. It is highest for compact, well-separated groups: the cluster model of K-means. Choosing between K-means and a density method by silhouette is therefore not a neutral test.

The bias is large on our data. On Wine, DBSCAN scores a higher silhouette than K-Means++ (0.42 against 0.28) while agreeing far less with the true classes (ARI 0.31 against 0.90). It earns that score by labelling 65% of the wines as noise, which the silhouette leaves out. On Ecoli, the silhouette ranks four methods in almost the reverse of their ARI order (Spearman  $\rho = -0.80$ ). [Lam: this Wine DBSCAN setting labels 64.6% of wines noise, above the agreed 50% cap (Section 3.3). Rerun the selection under the rule and update the numbers, or present it explicitly as what the silhouette

picks without the cap.] [Ky: which results CSV holds the Ecoli  $\rho = -0.80$ ?] On HDB resale flats, K-means on unscaled features scores 0.540 by splitting sales into four price bands, because price alone sets the distances.

The data can also mislead before any method runs. Treated as flat coordinates, latitude and longitude put 179.9°E and 179.9°W at opposite edges of the map, although they lie about 20 km apart. Storey numbers recorded in three-floor bands create thousands of identical values, which density methods read as dense clusters.

This report asks two questions. Which properties of a dataset decide whether a centroid, mixture, density or graph-based method succeeds? What can an analyst check without labels to avoid a misleading choice?

### 1.3 Contributions

[All: update dataset and method counts once Table 2 is final.]

- A comparison of four core methods (K-Means, K-Means++, DBSCAN, HDBSCAN) on eight datasets, extended with GMM, spectral and agglomerative clustering. Results are grouped by the data property that decides the outcome: cluster shape, density variation, dimensionality and the presence of gaps between groups (Section 3.4).
- Evidence from five datasets that the silhouette can prefer the wrong answer (Section 3.4.5), with a fake-data check that catches data containing no structure to find (Section 3.3). On HDB, only K-means beats data whose columns are shuffled independently (silhouette 0.317 against 0.218). Both density methods score below their fake-data baseline.
- Ablations that change one component at a time: initialisation, coordinate handling, feature scaling, dimensionality and values recorded in steps (Section 3.5). On Ecoli, DBSCAN and K-Means++ swap rank between three and four PCA dimensions.
- Seven implemented improvements, each tied to an observed failure and a measured effect (Section 3.6). Surface distance keeps together all 92 close earthquake pairs that straddle the 180° meridian; flat coordinates keep none. HDBSCAN keeps 62% of the sparse belts in the Americas, against 20% for DBSCAN with one global radius.
- Runtime on all 234,294 HDB sales (Section 3.7). K-means finishes in under 0.06 s and HDBSCAN in 212 s; DBSCAN would need about 6 GB for its neighbour lists. Section 3.9 turns these results into selection guidance.

### 1.4 Related work

**Centroid and mixture models.** Lloyd’s algorithm [12] minimises the within-cluster sum of squared distances by alternating assignment and centroid updates. It converges to a local optimum that depends on the starting centres. K-means++ [2] picks each starting centre with probability proportional to its squared distance from the centres already chosen, which gives an  $O(\log k)$  approximation to the optimum in expectation. A Gaussian mixture fitted by EM [7] generalises K-means to ellipsoidal clusters with soft assignments. Agglomerative clustering builds a hierarchy by repeatedly merging the closest pair of clusters. Under Ward’s criterion [18] the merge cost is the increase in within-cluster variance, the same objective as K-means.

**Density-based methods.** DBSCAN [8] defines a cluster as a maximal set of density-connected points. A point is a core point if at least `minPts` points lie within radius  $\epsilon$ . Points reachable from no core point are noise. The method needs no  $k$  and imposes no shape, but its single global  $\epsilon$  assumes all clusters have similar density. OPTICS [1] orders points by reachability so that clusters at several densities can be read from one plot. HDBSCAN [5] builds a hierarchy over mutual

reachability distances and keeps the clusters that persist longest. This removes  $\varepsilon$  and leaves a minimum cluster size as the main parameter.

**Graph-based methods.** Spectral clustering [14] embeds the points using the leading eigenvectors of a normalised affinity matrix, then runs K-means in that embedding. Groups connected through the affinity graph become compact in the embedding, so non-convex clusters can be separated. A dense affinity matrix needs  $O(n^2)$  memory, which limits the method to modest  $n$ .

**Validation and cluster tendency.** Internal indices, such as the silhouette [16], Davies–Bouldin [6] and Calinski–Harabasz [4] indices, score a partition from the data alone. External indices such as the Adjusted Rand Index [11] compare it with known labels. Von Luxburg et al. [13] argue that no criterion can judge a clustering apart from its intended use. A separate line of work tests whether data contain clusters at all. The Hopkins statistic [9] compares nearest-neighbour distances in the data with those in uniformly random points. The gap statistic [17] compares within-cluster dispersion with its value on reference data drawn from a null distribution. Our fake-data check applies the same idea to every method we compare. In high dimensions the ratio of nearest to farthest neighbour distance tends to one [3], which weakens any method built on a density threshold.

## 2 Methods

Owner: Ky (K-Means, K-Means++, DBSCAN, HDBSCAN); Lam (GMM, Spectral); Hoang Anh (Agglomerative)

### 2.1 K-Means and K-Means++

[Ky: Port Sec. 3.1.]

### 2.2 DBSCAN

[Ky: Port Sec. 3.2.]

### 2.3 HDBSCAN

[Ky: Port Sec. 3.3.]

### 2.4 Gaussian Mixture Model

[Lam: EM fit, covariance type used, parameters, complexity, assumptions.]

### 2.5 Spectral clustering

[Lam: Affinity graph, eigenvectors, K-means step; affinity used per dataset.]

### 2.6 Agglomerative clustering

[Hoang Anh: Linkage used and why; parameters; complexity.]

**Table 1:** Structural assumptions of the methods compared.

|                            | K-Means   | K-Means++ | DBSCAN          | HDBSCAN         | GMM      | Spectral | Agglom. |
|----------------------------|-----------|-----------|-----------------|-----------------|----------|----------|---------|
| Cluster shape assumed      | isotropic | isotropic | arbitrary       | arbitrary       | [Lam: ?] | [Lam: ?] | [HA: ?] |
| Requires $k$               | yes       | yes       | no              | no              | [Lam: ?] | [Lam: ?] | [HA: ?] |
| Explicit noise label       | no        | no        | yes             | yes             | [Lam: ?] | [Lam: ?] | [HA: ?] |
| Handles varying density    | no        | no        | no              | yes             | [Lam: ?] | [Lam: ?] | [HA: ?] |
| Parameter is interpretable | no        | no        | yes (km)        | partly          | [Lam: ?] | [Lam: ?] | [HA: ?] |
| Time complexity            | $O(nkdi)$ | $O(nkdi)$ | $O(n \log n)^a$ | $O(n \log n)^a$ | [Lam: ?] | [Lam: ?] | [HA: ?] |

<sup>a</sup> With an effective spatial index;  $O(n^2)$  worst case.

## 3 Experiments

### 3.1 Datasets

Owner: each dataset’s owner writes its paragraph (max. 120 words)

**Table 2:** Datasets, grouped by the data property each one tests. [DECIDE: final list; drop or keep Mall; Ky’s 1965–2016 earthquakes replaced by the 2023 surface-distance pipeline?]

| ID | Dataset                         | Owner          | $n$                 | $d$ | Labels        | Property tested               |
|----|---------------------------------|----------------|---------------------|-----|---------------|-------------------------------|
| D1 | make_moons (synthetic)          | Lam            | 1,000               | 2   | 2 classes     | curved shapes (control)       |
| D2 | Earthquakes, 2023, $M \geq 4.5$ | Wei Yew        | 7,643               | 2   | none          | long belts, uneven density    |
| D3 | Uber NYC pickups, Apr 2014      | Ky, Hoang Anh  | [DECIDE: 6k or 15k] | 2   | none          | density contrast in a city    |
| D4 | Ecoli                           | Ky             | 336                 | 7   | 8 classes     | dimensionality, rare classes  |
| D5 | Fashion-MNIST (PCA)             | Lam            | 3,000               | 50  | 10 classes    | high dimensions               |
| D6 | Wine                            | Lam, Hoang Anh | 178                 | 13  | 3 classes     | silhouette vs labels          |
| D7 | Breast cancer (Wisconsin)       | Hoang Anh      | 569                 | 30  | 2 classes     | silhouette vs labels          |
| D8 | HDB resale flats, 2017–2026     | Wei Yew        | 20,000 <sup>b</sup> | 4   | flat type (7) | no gaps, scale, recorded ties |

<sup>b</sup> Random sample of 234,294 usable sales; all 234,294 used for runtime (Section 3.7).

**D1 make\_moons.** [Lam: generator settings (noise 0.07), why it is the shape control.]

**D2 Earthquakes.** Every event of magnitude 4.5 or more in 2023 from the USGS ComCat catalogue [20]: 7,651 events, of which 8 volcanic eruptions near Saipan are removed, leaving 7,643 earthquakes. We cluster the epicentre only. Depth is left out because 44.2% of events carry the default depth of 10 km. Each method meets a different difficulty. Earthquakes lie along narrow belts thousands of kilometres long, which breaks the compact-cluster assumption of K-means. Density varies widely: the distance to the nearest earthquake runs from 1.2 km (5th percentile) to 100.5 km (95th), so no single DBSCAN radius suits every belt. And 17.2% of earthquakes lie within  $10^\circ$  of the  $180^\circ$  meridian, where flat coordinates cut the map. There are no labels.

**D3 Uber NYC.** [Ky + Hoang Anh: one merged paragraph; agree sample size and bounding box.]

**D4 Ecoli.** [Ky: Port from Sec. 4.1.]

**D5 Fashion-MNIST.** [Lam: 3,000-image sample, PCA to 50 dims; why subsampled.]

**D6 Wine.** [Lam + Hoang Anh: one merged paragraph; both runs give K-Means ARI 0.897.]

**D7 Breast cancer.** [Hoang Anh: 30 features, standardised; benign vs malignant.]

**D8 HDB resale.** Every resale of a public housing flat in Singapore from January 2017, from data.gov.sg [21]. The 6,051 sales after June 2026 (2.5%) have no price index value yet and are dropped, leaving 234,294. Methods are compared on a random sample of 20,000 (seed 0); all 234,294 are used for runtime. There are four features: floor area, remaining lease, storey, and price adjusted to 2026 levels with the HDB resale price index. Flat type (7 classes) is held out as a label, but it is nearly a function of size: floor area alone reaches ARI 0.601 against it. The data are hard for density methods. There are no gaps between groups of flats, and storeys are recorded in 3-storey bands, so 77.7% of sales share their floor area, lease and storey with at least one other sale.

## 3.2 Preprocessing

Owner: Wei Yew

Three rules apply wherever a dataset has the property they address. Each is tested by an ablation in Section 3.5.

**Coordinates on the sphere.** Latitude and longitude treated as flat numbers put two earthquakes 6.07 km apart at  $359.94^\circ$  of longitude from each other, because they lie on either side of the  $180^\circ$  meridian. Density methods therefore use the haversine (great-circle) distance on coordinates in radians. K-means needs a vector space to average in, so each epicentre becomes a 3D point on the unit sphere. The straight-line distance between two such points grows with their great-circle distance, so the clustering follows surface distance (Ablation B).

**Standardised features.** Features in their raw units are not comparable. On D8, price (in dollars) accounts for 100.0% of the squared distance between sales, so the other three features are ignored. Every feature is scaled to mean 0 and standard deviation 1 (Ablation C).

**Values recorded in steps.** When a feature is recorded in steps, many points share exact values, and density methods read each shared value as a dense spot. Before running a density method on D8, each stepped value is moved by a uniform random amount of up to half its step:  $\pm 1.5$  storeys,  $\pm 0.5 \text{ m}^2$  of floor area and  $\pm \frac{1}{24}$  year of lease. Price is continuous and is left unchanged. Results are reported with and without this jitter (Ablation E).

## 3.3 Evaluation metrics and parameter selection

Owner: Ky, Wei Yew

**Metrics.** Every result table reports the same columns. *Internal* scores need no labels. The silhouette coefficient [16] of a point is  $(b - a) / \max(a, b)$ , where  $a$  is its mean distance to its own cluster and  $b$  its mean distance to the nearest other cluster; we report the mean over points, from  $-1$  to  $1$ , higher is better. The Davies–Bouldin index [6] averages, over clusters, the worst ratio of within-cluster spread to between-centre distance; lower is better. Both are computed on clustered points only, so they say nothing about the points a density method calls noise. Two *coverage* figures therefore stand next to them: the noise fraction and the share of points in the largest cluster. *External* scores need labels. The Adjusted Rand Index (ARI) [11] counts pairs of points on which a clustering and the labels agree, corrected so that a random clustering scores about 0 and a perfect one 1. Normalised mutual information (NMI) measures the same agreement in terms of shared information. For both, noise counts as one more cluster, so a method cannot raise its score by discarding hard points. We also report the fit time.

**Fake-data check.** An internal score has no fixed scale: a silhouette of 0.3 can be strong on one dataset and meaningless on another. Each chosen setting is therefore run again on fake data that keeps each feature’s distribution but has no joint structure. For tables, each column is shuffled independently. For coordinates, points are drawn uniformly in the same latitude and longitude range. A result counts as structure only if the real data score clearly above the fake data.

**Selection rule.** One rule picks the setting of DBSCAN and HDBSCAN on every dataset: among the settings that give at least 2 clusters and label at most 50% of points as noise, take the one with the highest silhouette. The noise cap is needed because the silhouette ignores noise. Without it, the silhouette prefers settings that throw most of the data away. On D2 it would pick DBSCAN with  $\varepsilon = 20.4$  km, which scores 0.878 by calling 69.4% of earthquakes noise. On D8 it would pick HDBSCAN with `min_cluster_size` 400, which scores 0.448 with 70.9% of sales as noise. For K-means,  $K$  is chosen from the elbow in inertia, the silhouette, the gap to the fake-data silhouette and agreement between seeds; each dataset states which of these decided. On labelled data we report the rule’s choice and the setting with the best ARI, so the cost of choosing without labels is visible.

**Parameter grids.** K-means and K-means++:  $K = 2$  to 12, 10 starts each. DBSCAN: `minPts` (`min_samples` in scikit-learn) in  $\{5, 10, 20\}$ . Its radius  $\varepsilon$  comes from the k-distance curve, the sorted distance from each point to its `minPts`-th neighbour: on D2 we take the 25th, 50th, 75th, 90th and 95th percentiles of the curve (11 to 810 km); on D8 we take 0.5, 0.75, 1, 1.5 and 2 times the knee of the curve. HDBSCAN: `min_cluster_size` scaled to the data (25, 50, 100 on D2; 100, 200, 400 on D8) and `min_samples` in  $\{\text{default}, 5, 10\}$ . **[Ky, Lam, Hoang Anh: one line each for your datasets, and GMM components and spectral affinity.]**

**Reproducibility.** On D2 and D8 every random step uses seed 0, and repeated runs use seeds 0 to 19. They ran on Python 3.12.13, scikit-learn [15] 1.9.1 and NumPy 2.5.3 on an Apple M2 with 16 GB of memory. **[Ky, Lam, Hoang Anh: versions and hardware if different.]** **[DECIDE: shared code: Ky’s `src/core.py` or Wei Yew’s `metrics.py`, so every table has the same columns.]**

### 3.4 Comparison by data property

Owner: All

**[All: One results table per property (rows: dataset  $\times$  method; columns: #clusters, noise, S, DB, ARI, NMI, time, fake-data gap). Bold the best per dataset.]**

**Table 3:** D2 and D8 at the settings chosen in Section 3.3. S: silhouette; DB: Davies–Bouldin; null S: silhouette of the same setting on fake data (D2: shuffled coordinates for K-means, uniform points for density methods; D8: shuffled columns). “none”: no clusters formed on the fake data. D2 has no labels.

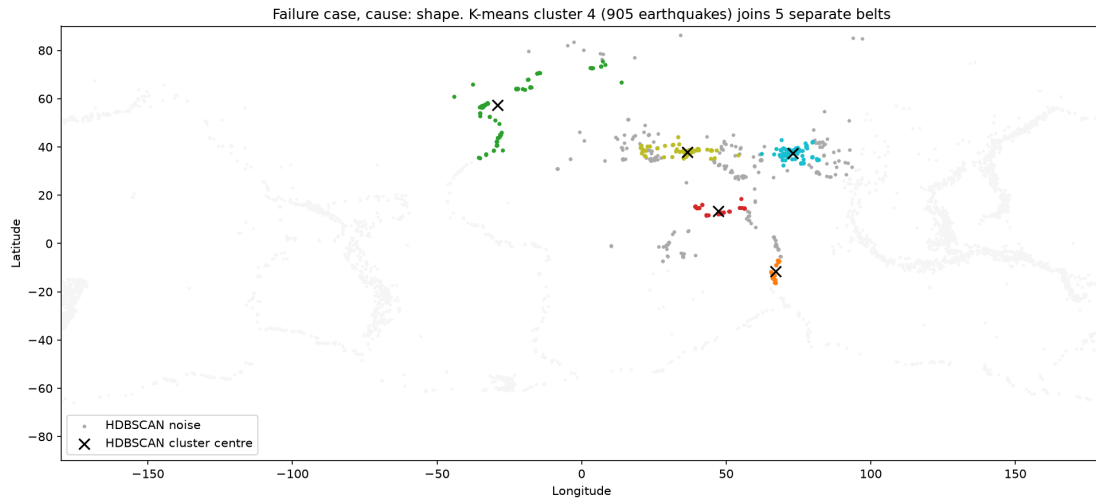
| Data | Method                | Clusters | Noise | Largest | S     | DB    | ARI   | NMI   | Time (s) | Null S |
|------|-----------------------|----------|-------|---------|-------|-------|-------|-------|----------|--------|
| D2   | K-Means ( $K = 6$ )   | 6        | 0%    | 44.3%   | 0.525 | 0.800 | –     | –     | 0.004    | 0.389  |
| D2   | K-Means++ ( $K = 6$ ) | 6        | 0%    | 31.0%   | 0.528 | 0.738 | –     | –     | 0.005    | 0.391  |
| D2   | DBSCAN (101.8 km, 20) | 41       | 40.4% | 11.8%   | 0.703 | 0.351 | –     | –     | 0.051    | none   |
| D2   | HDBSCAN (50)          | 28       | 30.7% | 7.5%    | 0.757 | 0.329 | –     | –     | 1.081    | 0.320  |
| D8   | K-Means ( $K = 4$ )   | 4        | 0%    | 36.2%   | 0.317 | 1.116 | 0.220 | 0.299 | 0.011    | 0.195  |
| D8   | K-Means++ ( $K = 4$ ) | 4        | 0%    | 35.1%   | 0.317 | 1.116 | 0.213 | 0.293 | 0.010    | 0.218  |
| D8   | DBSCAN (0.567, 10)    | 3        | 1.4%  | 97.1%   | 0.239 | 0.732 | 0.030 | 0.091 | 0.428    | 0.330  |
| D8   | HDBSCAN (200)         | 2        | 28.5% | 68.6%   | 0.160 | 1.133 | 0.072 | 0.101 | 2.022    | 0.593  |

#### 3.4.1 Curved and elongated clusters

Owner: Lam (D1), Wei Yew (D2)

[Lam: D1: DBSCAN ARI 0.982, Spectral 1.000, K-Means++ 0.481 (lam/results/make\_moons\_results.csv).]

K-means assumes compact clusters, and D2 breaks that assumption at planetary scale. Earthquakes follow plate boundaries, so a cluster is a long, narrow belt rather than a ball around a centre. K-Means++ with  $K = 6$  must cover the whole globe with six roughly round regions, so each region takes in several unrelated belts (Figure 1). One cluster of 905 earthquakes joins Turkey, the Reykjanes Ridge and Iceland, Afghanistan and Tajikistan, Somalia and Eritrea, and the Mid-Indian Ridge. HDBSCAN finds each of these as its own cluster, and their centres are 2,932 to 11,449 km apart. K-means also has no noise label, so every isolated earthquake in the oceans is assigned to some region. Its silhouette is still 0.528, well above the 0.391 of shuffled coordinates: the score confirms that the data are not random, not that the clusters are meaningful.



**Figure 1:** D2, failure of K-Means++ ( $K = 6$ ). All points shown in colour or dark grey belong to one K-means cluster; colours are the HDBSCAN clusters inside it, crosses their centres. Cause: cluster shape. Belts thousands of kilometres apart are joined because K-means can only draw compact regions.

### 3.4.2 Uneven density

Owner: Wei Yew (D2), Ky and Hoang Anh (D3)

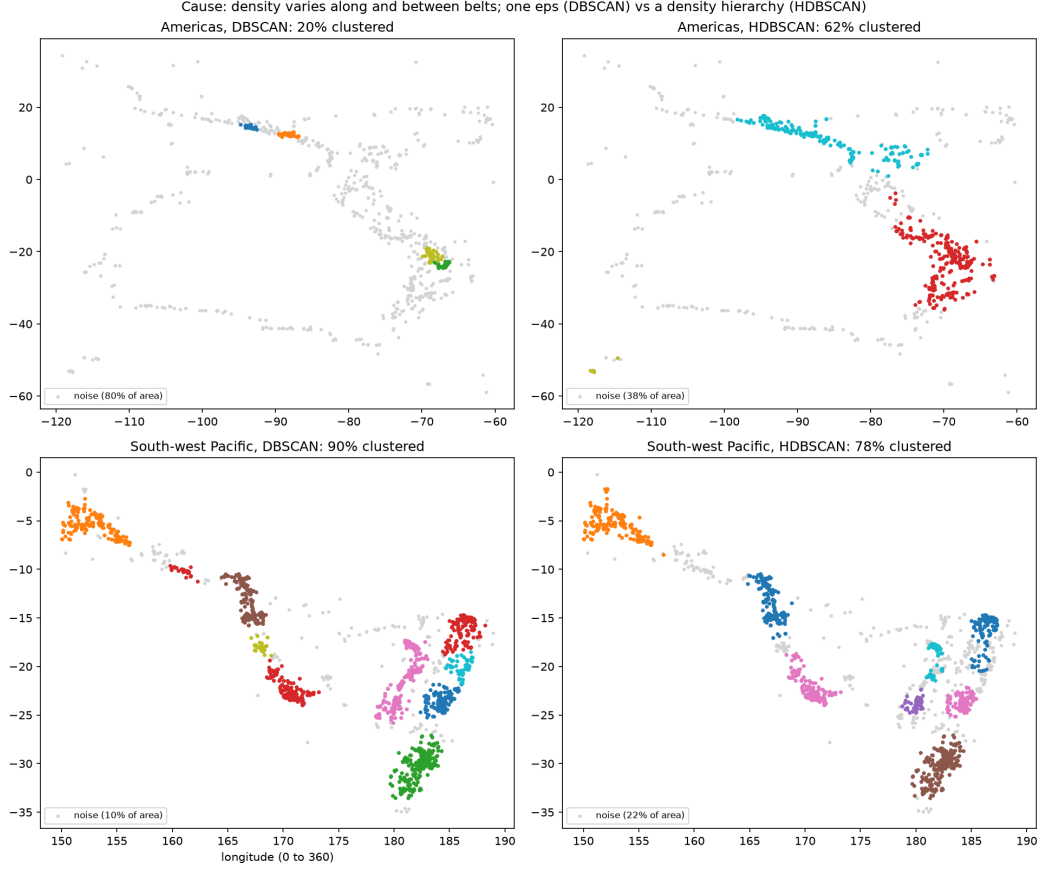
On D2 the distance to the nearest earthquake runs from 1.2 km to 100.5 km (5th to 95th percentile), and the k-distance curve rises smoothly with no knee. Any single  $\varepsilon$  is therefore too small for some belts and too large for others. At the chosen  $\varepsilon = 101.8$  km, DBSCAN clusters 90% of the earthquakes in the dense south-west Pacific but only 20% in the sparser Americas (Figure 2). HDBSCAN, which picks a density level per cluster, clusters 62% of the Americas and keeps the Mexico to Panama belt and the Chile and Argentina belt as long clusters. The price is paid in the dense region, where it clusters 78% instead of 90%. Overall HDBSCAN keeps more earthquakes (69.3% against 59.6%) with a higher silhouette (0.757 against 0.703, Table 3).

[Ky + Hoang Anh: D3: Midtown vs outer boroughs; knee  $\varepsilon$  vs silhouette-optimal  $\varepsilon$ .]

### 3.4.3 High dimensions

Owner: Ky (D4), Lam (D5)

[Ky: D4: crossover between 3 and 4 PCA dimensions (Ablation D).] [Lam: D5: DBSCAN ARI 0.033; HDBSCAN labels 71% of points noise.]



**Figure 2:** D2, DBSCAN ( $\epsilon = 101.8$  km) against HDBSCAN in a sparse region (top, the Americas) and a dense one (bottom, the south-west Pacific); grey points are noise. Success of HDBSCAN on the sparse belts, at the cost of trimming the dense ones. Cause: density varies along and between belts.

#### 3.4.4 No gaps between groups

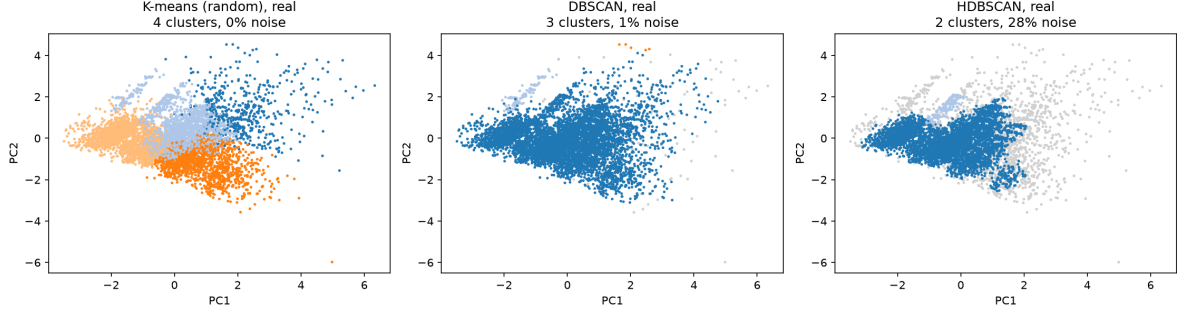
Owner: Wei Yew (D8)

D8 is one continuous mass: flats come in every combination of size, lease, storey and price, with no empty space between groups. Table 3 shows what each method does with it. K-means still finds a weak partition, four market segments (newer mid-size, older small, large older, newer high-floor), with a silhouette of 0.317 against 0.218 on shuffled columns. That ratio of 1.45 is modest, but it is the only result on D8 that beats its fake data. Density methods have no sparse region to cut along. The DBSCAN setting chosen by the rule puts 97.1% of sales in one cluster, and HDBSCAN finds one core holding 68.6% of sales plus a compact block of new 3-room flats, and calls 28.5% noise. Both score below the same setting on shuffled columns (0.239 against 0.330, and 0.160 against 0.593). On shuffled columns HDBSCAN calls 94.0% of sales noise and keeps two small, well-separated clusters, which the silhouette rewards. On data with no gaps, a density method has nothing to find, and the selection rule cannot tell.

#### 3.4.5 Silhouette versus ground truth

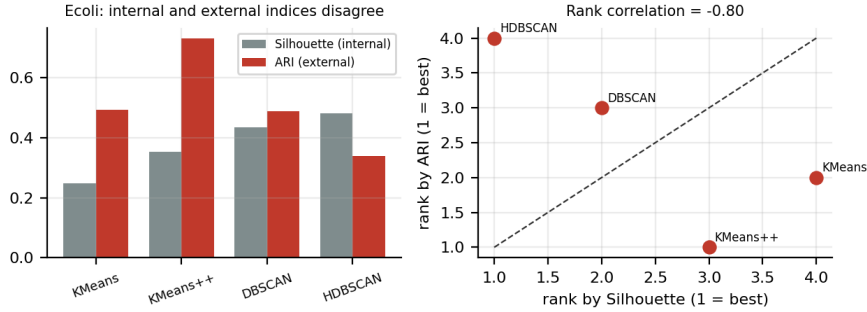
Owner: All

[All: One paragraph, one line per dataset: D4 best silhouette (HDBSCAN) has the worst ARI, rank correlation  $-0.80$  (Ky); D6 DBSCAN silhouette 0.42 vs K-Means++ 0.28, ARI 0.31 vs 0.90 (Lam); D7 K-Means top silhouette 0.343, GMM best ARI 0.774 vs 0.654 (Hoang Anh); D8 unscaled K-means silhouette 0.540 from price bands, against 0.317 standardised (Wei Yew, written in Ablation C); D2 K-means silhouette flat at 0.516 to 0.532 for every  $K$  from 4 to 12, so



**Figure 3:** D8 at the chosen settings on recorded data, drawn on the first two principal components (6,000 sales; grey: noise). K-means cuts the mass into four segments; DBSCAN and HDBSCAN return one dominant cluster. Cause: no gaps between groups.

it cannot choose  $K$  (Wei Yew, [weiyew/sc4020-earthquake/results/kmeans\\_k\\_selection.csv](https://github.com/weiw/sc4020-earthquake/tree/main/results/kmeans_k_selection.csv)); D5 HDBSCAN top silhouette with 71% noise (Lam).]



**Figure 4:** [Ky: existing figure; extend to D5–D8 or make one panel per dataset.]

### 3.5 Ablation studies

Owner: Ky, Wei Yew

#### 3.5.1 A: Initialisation (K-Means vs K-Means++)

[Ky: Port Ablation A (30 seeds, D3/D4). Add ARI mean  $\pm$  s.d. over the 30 seeds: the 0.24 ARI claim currently rests on seed 42 only.]

On D2 and D8 each initialisation ran from one start with 20 seeds at the chosen  $K$ ; we measure how far each run’s inertia lies above the best run. On D2 ( $K = 6$ ) K-Means++ helps clearly: the mean gap falls from 10.9% to 4.8%, the worst from 18.8% to 11.5%, and iterations from 15.3 to 10.3. It does not remove the need for restarts. Two K-Means++ runs still agree only at ARI 0.80 on average, against 0.78 for random starts, because the belts give many local optima of similar inertia. On D8 ( $K = 4$ ) the choice does not matter: all 40 runs end within 0.009% of the best inertia. Initialisation matters when the objective has many local optima of different quality; on D8 every start reaches practically the same one.

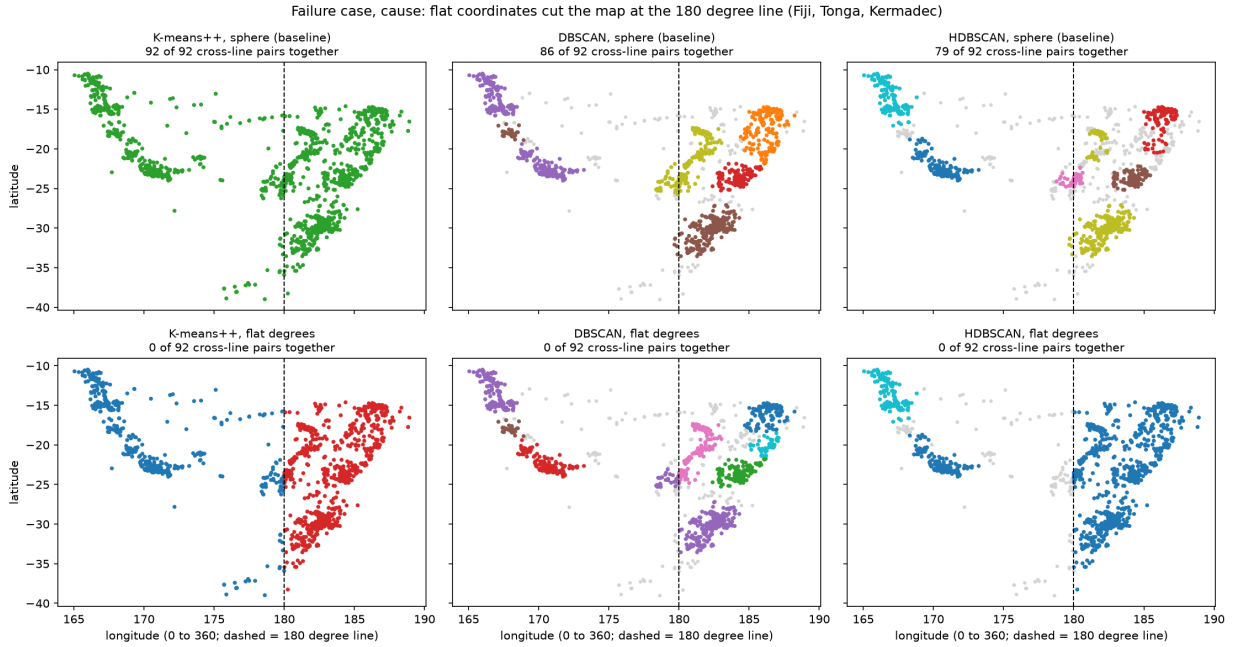
#### 3.5.2 B: Distance on geographic coordinates

Every method ran at its chosen setting twice on D2: once on the sphere (Section 3.2) and once on latitude and longitude as flat numbers (DBSCAN’s  $\epsilon$  converted at 111.2 km per degree). To isolate the effect of the 180° meridian we count *cross-line pairs*: the 92 pairs of earthquakes within 50 km of each other on opposite sides of the line. Table 4 shows that flat coordinates separate every

one of these pairs, for every method (Figure 5). The Fiji–Tonga–Kermadec system is cut in two. K-Means++ changes everywhere: its silhouette falls from 0.528 to 0.404 and its partition agrees with the sphere partition at ARI 0.557 only. DBSCAN hides the failure in its global numbers. Its silhouette moves from 0.703 to 0.701 and its partition still agrees at ARI 0.948, because only the clusters that cross the line are split. A check aimed at the known problem, here the cross-line count, finds what the internal scores miss.

**Table 4:** D2 on the sphere and on flat degrees. Cross-line pairs: 92 close pairs that straddle the 180° meridian; “kept” counts pairs placed in the same cluster.

| Method    | Distance | Clusters | Silhouette | ARI vs sphere | Pairs kept |
|-----------|----------|----------|------------|---------------|------------|
| K-Means++ | sphere   | 6        | 0.528      | 1.000         | 92 of 92   |
|           | flat     | 6        | 0.404      | 0.557         | 0 of 92    |
| DBSCAN    | sphere   | 41       | 0.703      | 1.000         | 86 of 92   |
|           | flat     | 43       | 0.701      | 0.948         | 0 of 92    |
| HDBSCAN   | sphere   | 28       | 0.757      | 1.000         | 79 of 92   |
|           | flat     | 23       | 0.721      | 0.784         | 0 of 92    |



**Figure 5:** Fiji, Tonga and Kermadec, drawn on longitudes 0 to 360° (dashed: the 180° meridian). Top: on the sphere, clusters cross the line. Bottom: on flat degrees, every method splits the system at the line. Cause: flat coordinates treat 179.9°E and 179.9°W as 359.8° apart.

[**DECIDE:** Ky’s projected-km pipeline has the same  $\pm 180^\circ$  cut. Rerun D3 on the sphere, or state that city scale makes it irrelevant.]

### 3.5.3 C: Feature scaling

[**Ky:** D4: standardisation lifts K-Means++ ARI 0.497  $\rightarrow$  0.731; DBSCAN unaffected once  $\varepsilon$  is re-tuned.]

On D8, K-means ( $K = 4$ ) on raw units scores a silhouette of 0.540, well above the 0.317 of the standardised run. The higher score is not a better clustering. In dollars, price makes up 100.0% of the squared distance between sales, so the four clusters are price bands and floor area, lease and storey play no part. The raw partition agrees with the standardised one at ARI 0.261 only, and

neither matches flat type (ARI 0.229 raw, 0.214 standardised). Scaling decides which question the clustering answers. The silhouette, measured in each run’s own units, cannot say which is right.

### 3.5.4 D: Dimensionality

[Ky: Port Ablation D. Call it PCA dimension; “intrinsic dimensionality” overclaims.]

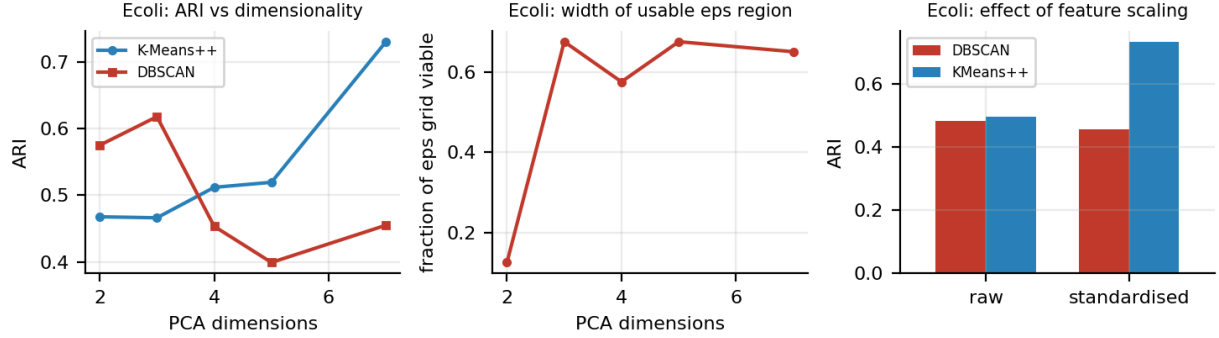


Figure 6: [Ky: existing figure (Ablations C and D on D4).]

### 3.5.5 E: Values recorded in steps

On D8 the selection rule picks density settings dominated by one cluster: it holds 97.1% of sales for DBSCAN, and 68.6% for HDBSCAN, which calls another 28.5% noise (Section 3.4.4). A natural repair is to add a cap on the largest cluster. We tested caps from 90% down to 50% on the data as recorded and on jittered data (Section 3.2). On recorded data a cap of 50% does give balanced partitions, but of the wrong kind: all 27 DBSCAN clusters, and 6 of the 7 HDBSCAN clusters, each contain a single storey band (Figure 7). They are stacks of identical recorded values, not groups of similar flats. On jittered data, where those ties are spread out, *no* setting meets a cap of 50% or 60%. The balanced partitions existed only because of the recording steps. At the uncapped HDBSCAN setting, by contrast, recorded and jittered data agree (ARI 0.886), so the chosen setting is not a tie artefact. Two lessons follow. Stepped values must be checked before a density method runs. And a constraint added to a selection rule can make it pick artefacts, which is why we kept the rule unchanged.

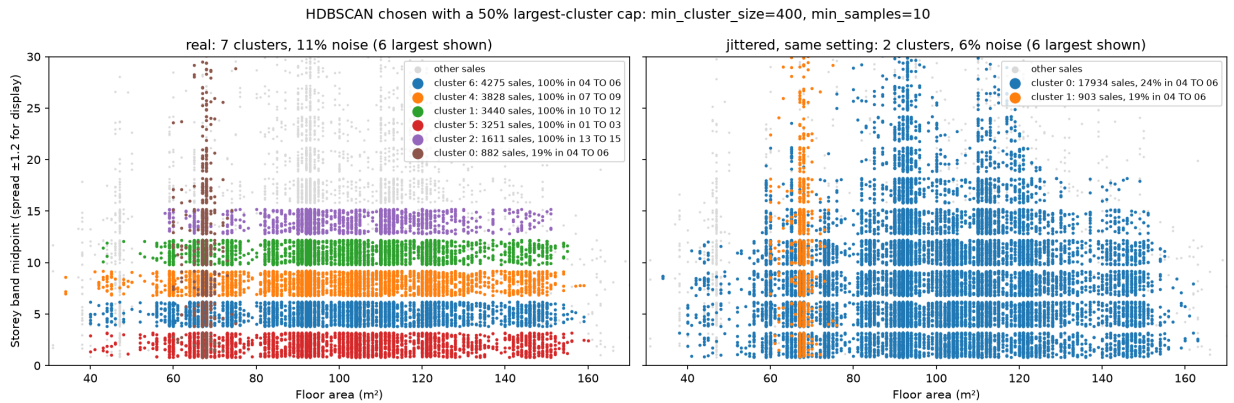


Figure 7: D8, HDBSCAN chosen under a 50% largest-cluster cap. Left, recorded values: 6 of 7 clusters are single storey bands. Right, the same setting on jittered values: 2 clusters. Cause: storeys recorded in 3-storey bands create exact ties that density methods read as dense regions.

### 3.5.6 F: Data subsets

Each D2 subset is compared with a random subset of the same size, so that an effect of the subset is not confused with an effect of size. Clustering January to June or July to December separately, K-Means++ reproduces its full-year labels at ARI 0.963 and 0.829 (random halves: 0.968 and 0.994); the drop for the second half comes from aftershock sequences. Density methods change as much on random halves as on the real ones (HDBSCAN 0.414 and 0.472, against 0.505 and 0.544). Their parameters are counts of points, so their output depends on how many points there are. Keeping only magnitude 5.0 and above (1,780 earthquakes) behaves like a random subset of the same size (DBSCAN ARI 0.343 against 0.315, HDBSCAN 0.322 against 0.303). Larger earthquakes do not outline the belts more clearly, so we keep the 4.5 threshold.

## 3.6 Improvements to the methods

Owner: All

**Table 5:** Implemented improvements. Each fix targets a failure observed in Section 3.4.

| Failure                   | Cause                 | Fix                           | Data   | Before → after                        | Owner       |
|---------------------------|-----------------------|-------------------------------|--------|---------------------------------------|-------------|
| Clusters split at 180°    | flat lat/lon          | surface distance              | D2     | 0 → 92 of 92 pairs kept               | Wei Yew     |
| Poor local optimum        | random seeding        | K-Means++                     | D4     | ARI 0.493 → 0.731                     | Ky          |
| Sparse belts lost         | one global $\epsilon$ | HDBSCAN                       | D2     | Americas kept 20% → 62%               | Wei Yew     |
| One feature dominates     | unequal scales        | standardisation               | D4     | ARI 0.497 → 0.731                     | Ky          |
| Clusters are storey bands | recorded steps        | jitter within step            | D8     | band clusters 107 of 124 → 0 of 23    | Wei Yew     |
| Curved clusters cut       | convex assumption     | spectral clustering           | D1     | ARI 0.481 → 1.000                     | Lam         |
| Degenerate $k$ chosen     | silhouette alone      | constraints + fake-data check | D2, D3 | D2 null gap 0.015 → 0.144; D3 [Ky: ?] | Ky, Wei Yew |

**Surface distance (D2).** Haversine distance for density methods and 3D unit vectors for K-means measure what matters, the distance along the Earth’s surface. The fix is exact, and its only cost is that  $\epsilon$  must be given in kilometres and the coordinates converted. It changes the answer for every method (Ablation B), which makes it a correctness fix rather than a tuning choice.

**HDBSCAN instead of DBSCAN (D2).** HDBSCAN replaces the single radius with a hierarchy of densities and keeps the clusters that persist longest, so sparse and dense belts can both survive. In the Americas it clusters 62% of earthquakes against 20%. The costs are real: dense cores are trimmed (78% against 90% in the south-west Pacific), each fit takes about 20 times longer (1.08 s against 0.05 s), and `min_cluster_size` is a count, so the result changes with the size of the dataset (Ablation F).

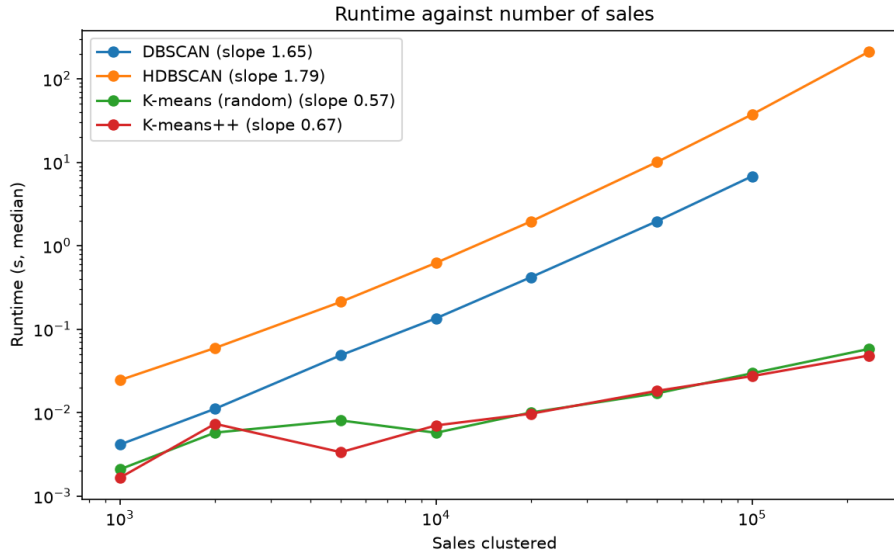
**Jitter within the recording step (D8).** Moving each stepped value by up to half its step spreads exact ties without moving any sale into another band. Across the 9 HDBSCAN settings, 107 of the 124 clusters found on recorded data are single storey bands; on jittered data, 0 of 23 are. The fix costs nothing at the chosen setting (recorded and jittered agree at ARI 0.886), but it adds randomness, so the seed must be fixed and reported. It also shows that D8 has fewer real density clusters than the recorded data suggest.

**Fake-data check for choosing  $K$  (D2).** The silhouette alone picks  $K = 2$  on D2 (0.544, the highest of any  $K$ ), but shuffled coordinates score 0.529 at the same  $K$ : the split follows the distribution of latitude and longitude on their own, not their combination. Requiring a clear gap over the fake data rules out  $K = 2$  and  $K = 3$ . From  $K = 4$  the gap is 0.126 to 0.149 and almost flat, so the final choice,  $K = 6$ , rests on agreement between seeds (ARI 0.993). The check costs extra clustering runs on fake data for each  $K$ . [All: Proposed but not implemented: e.g. ground truth for D2 from plate-boundary maps; OPTICS; approximate spectral clustering for large  $n$ .]

### 3.7 Runtime and scalability

Owner: Wei Yew, Lam

D8 is large enough to measure how each method grows. Each method ran at its chosen setting on random subsets of 1,000 to 234,294 sales (K-means from one start). Figure 8 plots the median time against size on log scales, where the slope of a line estimates the exponent  $b$  in  $t \propto n^b$ . K-means is far cheaper than either density method at every size: under 0.06s on all 234,294 sales. Its fitted slopes (0.57 and 0.67) are below 1 only because a fixed start-up cost dominates at these sizes; each iteration is linear in  $n$ . HDBSCAN grows with slope 1.79, from 0.02s at 1,000 sales to 212s at 234,294. DBSCAN grows with slope 1.65, but at full size it was not run at all, because of memory rather than time. With a fixed  $\varepsilon$ , every sale has more neighbours as the data grow. Estimated from 2,000 probe points, the full data would hold about 742 million neighbour pairs, about 6 GB for the neighbour indices alone, which scikit-learn’s DBSCAN stores in full. HDBSCAN, slower at every size, did run on all sales within 16 GB. [Lam: Why D5 needed subsampling for Spectral, DBSCAN and HDBSCAN; fit times per method.] [Ky: Existing Fig. 9 runtime at  $n \leq 23,412$ , if kept.]



**Figure 8:** D8, median fit time against the number of sales, log–log, at the chosen settings; the slope of each line is in the legend. DBSCAN stops at 100,000 sales: the full data would need about 6 GB of neighbour indices.

### 3.8 Success and failure cases

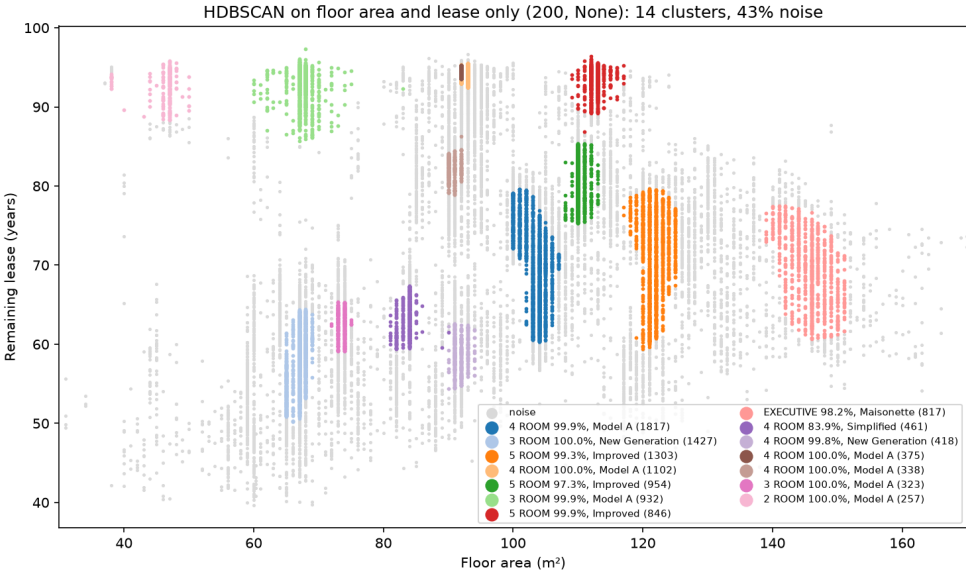
Owner: each owner: one figure per case, with the cause named in the caption

#### 3.8.1 Successes

**HDBSCAN on D2: belts of varying density.** Shown in Figure 2. HDBSCAN keeps the sparse belts of the Americas (62% of earthquakes clustered, against 20% for DBSCAN) as long clusters that follow the plate boundary. It succeeds because its density threshold adapts to each cluster, the property D2 needs most. [Lam: DBSCAN / Spectral on D1.] [Ky: K-Means++ on D4: compact classes in 7 dimensions.]

**HDBSCAN on D8, floor area and lease only: flat designs.** This feature set was chosen after seeing the main results, so we present it as an ablation finding and a sanity check, not as a tuned setting. Dropped to two features, D8 does contain dense spots: standard flat designs, built

to fixed plans in particular years. HDBSCAN finds 14 clusters (43.2% noise), and 13 of them are at least 97% one flat type (Figure 9); in the median cluster one flat model accounts for 83% of sales. A simple grouping by flat model and lease would give similar groups, which is the point: the method recovers a structure we can name. ARI against flat type is still only 0.127, because each type is split into several designs and the noise counts as a cluster. A low ARI here reflects a finer partition than the labels, not a failure.



**Figure 9:** D8, success of HDBSCAN on floor area and remaining lease only (`min_cluster_size` 200; grey: noise). Each cluster is one standard flat design; the legend gives its main flat type and model. Cause of success: fixed building plans create real dense spots in these two features.

### 3.8.2 Failures

**K-Means++ on D2: unrelated belts merged.** Shown in Figure 1. One cluster joins five belts whose centres are 2,932 to 11,449 km apart, from Iceland to the Mid-Indian Ridge. Cause: cluster shape. Compact regions cannot follow long belts.

**Density methods on D8: one mass, or storey bands.** Shown in Figures 3 and 7. At the chosen settings DBSCAN puts 97.1% of sales in one cluster. Under a cap on the largest cluster, the clusters become single storey bands. Causes: no gaps between groups, and values recorded in steps. [Lam: DBSCAN on D5.] [Ky: HDBSCAN on D4: 37% noise removes rare classes.]

All: success/failure panel grid: one panel per case above

**Figure 10:** [All: caption naming the cause of each case]

### 3.9 Key factors and practical guidance

Owner: Ky

[**Ky**: Port Sec. 7.3 (key factors) and re-rank them with evidence from all datasets: cluster shape, density variation, dimensionality, gaps between groups, scaling, initialisation, recorded steps, parameter choice.] [**Ky**: Port Sec. 7.4 (guidance). Add: compare every result against fake data (Wei Yew); check for values recorded in steps (Wei Yew); report noise % next to every silhouette.]

## 4 Conclusion

Owner: Ky

[**Ky**: Summary of findings with numbers; answer the title question.]

## Limitations

[**All**: One line each: D2 has no ground truth; D8 has no location feature; oracle ARI on labelled data is optimistic; datasets up to 50 dimensions.]

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- [**Ky + Hoang Anh:** Uber source.] [**Lam + Hoang Anh:** Wine, Mall.] [**Hoang Anh:** Breast cancer.]